

- DevOps is not a framework
- DevOps is not a set of design patterns, where problems and solutions are well defined
- DevOps is not scripting
- DevOps is not automation only

Dev	Ops
Maintain code quality and standards	Automated provisioning and de-provisioning of environments
Write robust applications using design patterns and best practices	Scaling up/down, in/out
Write practical unit tests and utilise active test-driven development	Manage Infrastructure as a Code (IaC) to maintain version and visibility
Deployment to the local environment	Configuration management
Verifying behaviour of an application	Monitoring and measurements
The need for communication and collaboration of Dev, test, security, and Ops (cloud/containers) teams.	

Table 1: Responsibilities of Dev and Ops teams

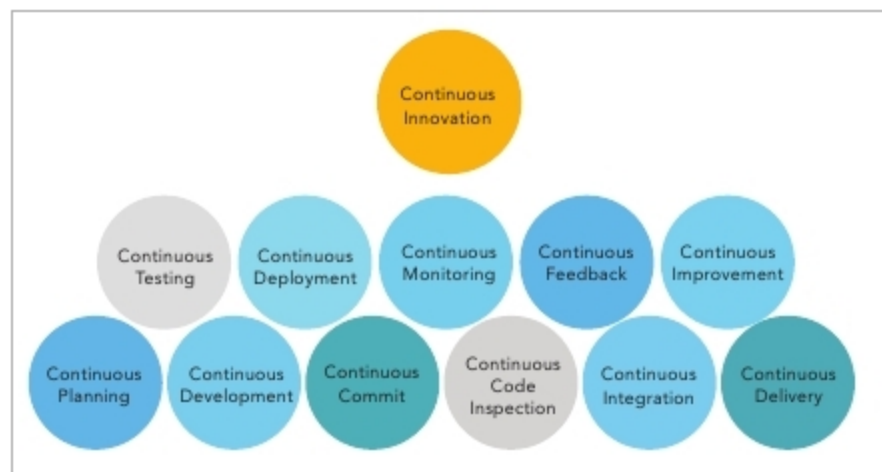


Figure 1: Continuous practices

Let's also understand the business and technical benefits realised by organisations after a cultural transformation, by using DevOps practices (Table 2).

Business benefits	Technical benefits
Improved communication and collaboration between teams	Early detection of failures
Happy people	High availability
Faster time to market	Stable and consistent environment
Quality releases	Efficient utilisation of resources
Productivity gains	Easy rollback: Enhanced recovery time
Transparency in process execution	Single-click deployment
Measurement of achievements; for example, improvement in deployment time	Continuous deployment
Agile teams	Monitoring
Reliability and reusability	Cross-skills learning

Table 2: Benefits of DevOps

In this article, we will get hands-on understanding of Azure DevOps using some concepts, and then implement CICD for the sample application.

Implementation of DevOps practices

Continuous practices play an important role in culture transformation. Figure 1 shows some, if not all, continuous practices.

Let's understand some continuous practices in detail, as given in the table below.

Continuous planning	Agile is an idea or a philosophy that brings pace to development and delivery. DevOps practices make sure that this pace is maintained, and hence they complement agile. There are many frameworks that come under the agile umbrella. Scrum is one of the most popular frameworks in the market. Following are the benefits of implementing a continuous planning practice in the organisation: • Visibility • Better collaboration and communication • Continuous feedback • High quality • Faster time to market • Focus on quality, end users and productivity gains • Predictability and room for change The tools related to continuous planning are: • Azure DevOps • ServiceNow • Atlassian Jira
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